Ex - 07
Data Visualization Using Power BI

AIM

To visualize data using Power BI by connecting data sources, creating charts, applying DAX calculations, and building an interactive dashboard using the T20 Cricket dataset.

PROBLEM STATEMENT

To analyze the T20 Cricket dataset using Power BI by creating bar charts, line charts, pie charts, calculated columns, and measures to present meaningful insights such as total runs, top batsmen, and matches won by each team.

PROCEDURE

1. Open Microsoft Power BI Desktop and explore the Report View, Data View, Model View, and Visualizations pane.
2. Connect to the T20 Cricket dataset (Excel or CSV) using Home > Get Data, select the required file, and click Load to import the data.
3. Connect to a SQL database by selecting Home > Get Data > SQL Server, entering the server details, and selecting the required tables.
4. Open Power Query Editor using Transform Data to remove duplicates, handle missing values, rename columns, and change data types.
5. Create a bar chart by selecting the Clustered Bar Chart visual and dragging batsmanName to the Y-axis and runs to the X-axis to show runs scored by each batsman.
6. Create a line chart by selecting the Line Chart visual, placing the match date field on the X-axis and runs on the Y-axis to analyze the trend of runs by match date.
7. Create a pie chart by selecting the Pie Chart visual, placing winner in the Legend and the count of match_id in Values to display the distribution of match wins.
8. Create a calculated column using Table tools > New column and enter a DAX formula to calculate additional values for each row.
9. Create measures using Table tools > New measure and apply DAX functions such as SUM(), DISTINCTCOUNT() and DIVIDE() to calculate totals and averages.
10. Build an interactive dashboard-style report by arranging charts, KPI cards, and slicers, formatting titles and colors, testing filters, and saving the report as a .pbix file.

IMPLEMENTATION

1. Calculated Column (DAX)

```
Run Category = IF(Cricket[runs] >= 50, "50+ Runs", "Below 50")
```

2. Measure - Total Runs

```
Total Runs = SUM(Cricket[runs])
```

3. Measure - Total Players

```
Total Players = DISTINCTCOUNT(Cricket[batsmanName])
```

4. Measure - Total Matches

```
Total Matches = DISTINCTCOUNT(Cricket[match_id])
```

Dashboard components:

• Cards - Total Runs, Total Players, Total Matches

• Bar chart - Batsman runs (Sum of runs by batsmanName)

• Column chart - Matches won by team (Count of match_id by winner)

• Line chart - Runs by match date

• Pie chart - Match wins distribution (Count of match_id by winner)

• Slicer - Filter by team or batsman

Note: Replace the table and column names with those in your dataset.

OUTPUT



Fig: T20 Cricket Analytics Dashboard (Power BI)

RESULT

Data was successfully visualized using Power BI on the T20 Cricket dataset. Bar charts, line charts, pie charts, DAX calculated columns, and measures were created and combined into an interactive dashboard for effective data analysis.